Republic of the Philippines
Iloilo Science and Technology University
Burgos St., La Paz, Iloilo City, 5000 Philippines
Trunkline: (+6333) 320-7190 | Telefax: (+6333) 329-4274
https://www.isatu.edu.ph/
mail@isatu.edu.ph

BACONG PILIPINAS

CODELYMPICS 2026

STUDENT INNOVATORS EDITION

PROJECT CONCEPT

PROJECT TITLE
BAHAWATCH - Community Flood & Road Condition Information App

DESCRIPTION

A community-powered app that helps users quickly view recent, location-specific reports about flooding and road conditions for specific roads and areas. The initial MVP is a functional mobile app prototype that organizes reports by location, road condition, severity, time, freshness, and verification status.

PROJECT CONCEPT

During heavy rainfall, people may know that bad weather is occurring without knowing the actual condition of the particular road or area they need to use. Existing sources such as weather applications, social media, news, and official announcements provide valuable information, but they do not always answer the immediate ground-level question: “What has recently been reported about this specific road?”

BAHAWATCH addresses this information gap through a community-powered app that turns scattered community observations into structured, road-specific reports. A person who encounters flooding can safely submit a report containing a map location, road condition, severity, time, and optional photo or video. Other users can select a specific road and see its latest available reports, freshness, verification status, and changes in reported condition over time.

The app does not predict floods or replace official warnings. It provides a clearer way to find recent community-reported ground information for a specific road or area. This information may help users make better-informed decisions, but BAHAWATCH does not tell users whether they should travel.

THE MAIN PROBLEM

The problem is not the lack of weather information. The problem is the gap between general weather information and specific, recent information about actual road and area conditions.

- Weather applications may show forecasts and rainfall conditions.
- Social media can contain useful reports, but information is scattered across posts, pages, and group chats.
- News and government announcements may provide warnings or advisories, but may not cover every road or location.
- Maps can show where places are, but do not necessarily show recent community-reported flood conditions.

BAHAWATCH focuses on organizing available ground reports by LOCATION, TIME, SEVERITY, ROAD CONDITION, FRESHNESS, and STATUS so users can quickly understand what has recently been reported about a specific road or place.

THE INFORMATION GAP

LOCATION-SPECIFIC INFORMATION PEOPLE MAY NEED

- Is the road I need to use flooded?
- How severe is the reported flooding?
- Is the road passable?
- When was the report submitted?
- Has someone verified or updated it?
- Has the area already been cleared?

The app is designed to make this second type of information easier to find without claiming complete live coverage.

PROJECT OBJECTIVE

Main Objective

To provide an easier and more trustworthy way for people to access recent, location-specific information about reported flood and road conditions for a particular road or area through a community-based map.

Specific Objectives

- Allow people to submit flood and road-condition reports when they are in a safe location.
- Capture location, road condition, severity, submission time, and optional photo/video.
- Display available reports on an interactive map.
- Make the freshness of each report immediately visible.
- Distinguish submitted reports from reviewed or verified reports.
- Allow reports to be updated or marked as cleared.
- Reduce the difficulty of finding scattered flood information across different sources.
- Validate whether the app is useful to residents and commuters as primary users, and to motorists, delivery riders, businesses, and potential institutional partners as secondary users.

WHO EXPERIENCES THE PROBLEM?

Residents

People living in flood-prone areas may need to know the condition of nearby roads and locations.

Commuters

Students and workers travelling during or after heavy rainfall may need information before passing through affected areas.

Drivers and Delivery Riders

People who need to travel through specific roads may benefit from knowing whether a road has recently been reported as passable, difficult, or not passable.

Local Businesses

Flooding can affect customers, employees, deliveries, and daily operations.

LGUs and Disaster Response Groups

To provide an easier and more trustworthy way for people to access recent, location-specific information about reported flood and road conditions for a particular road or area through a community-powered map app.

HOW THE CONCEPT WORKS

- A person observes flooding or a road condition.
- The person moves to or remains in a safe location before using the app.
- The user submits the location, road condition, severity, time, and optional photo/video.
- The app records the report and places it on the map.
- The report receives a freshness and review status.
- An administrator or authorized reviewer can review and verify the report.
- Other users can view the report and its details.
- Reduce the difficulty of finding scattered flood information across different sources.

The app does not make travel decisions for users or recommend a route. It provides structured information so users can judge the available evidence themselves.

FLOOD REPORT

Location

The report is linked to a map location, with the option to identify the road, barangay, or nearby landmark.

Road Condition

- Passable
- Difficult to Pass
- Not Passable
- Cleared

Severity
- Low
- Moderate
- High

For the initial MVP, the categories will be Passable, Difficult to Pass, Not Passable, and Cleared. These definitions will be tested with users and refined if needed.

Time
The app records the date and time when the report was submitted.

Photo/Video
A photo or video may be attached when available and when obtaining it can be done safely.

Report Status
- Submitted
- Under Review
- Verified
- Needs Update
- Cleared

REPORT FRESHNESS AND TRUST

Flood conditions can change quickly. Therefore, the app should make report age visible instead of presenting every report as a current fact.

- Reported 8 minutes ago
- Reported 45 minutes ago
- Needs Update
- Verified
- Cleared

Reports are not automatically treated as verified information. A community report starts as unverified and may later be reviewed by an administrator or authorized reviewer. Users can always see the report's status and age.

Where possible, the app can use supporting indicators such as photo evidence, recent timestamps, duplicate reports, and verification status to help users judge the reliability of information.

SAFETY CONSIDERATION

Safety is a core requirement of the app. The platform should never encourage users to approach floodwater or dangerous areas to submit information.

- Do not enter floodwater to take a photo.
- Do not stop in a dangerous location just to submit a report.
- Do not use the website while driving.
- Do not remain in an unsafe area to provide an update.
- Submit reports only from a safe location.
- The reporting form should include a short safety reminder before submission.

IS IT REALLY REAL-TIME?

The app should not claim complete real-time coverage. Instead, BAHAWATCH provides recent, time-stamped reports and clearly shows when each report was submitted.

For example: “Reported 12 minutes ago.” This allows users to judge the freshness of the information themselves. If no recent report exists for an area, the app should not imply that the area is clear.

WHAT IF NOBODY REPORTS?

The app cannot automatically know that an area is flooded when there is no available report. For the MVP, community reporting is the primary information source.

Future expansion may include contributions from LGUs, disaster response groups, authorized community members, and community organizations. These sources can increase coverage while keeping the distinction between community reports and official information.

EXISTING ALTERNATIVES AND COMPETITOR VALIDATION

The team recognizes that flood and road information can already be shared through social media, messaging platforms, navigation applications, and government disaster-apps. BAHAWATCH therefore does not claim to be the first system for reporting floods or road conditions.

Instead, the team will compare BAHAWATCH with existing alternatives during validation to determine whether users still experience a specific road-condition information gap and whether the road-specific view, freshness indicators, verification context, and condition timeline provide meaningful additional value.

The key question is not "Does any flood-reporting system already exist?" The key question is "What information do people still struggle to find when they need to understand the recent reported condition of one specific road?"

WHAT MAKES THE IDEA DIFFERENT?

The innovation is not simply allowing people to report floods. People can already post flood observations on social media, messaging platforms, and other services. BAHAWATCH focuses on the road-specific experience: organizing recent observations around one location so users can quickly compare the latest condition, timestamp, verification status, and changes over time.

- LOCATION - Where exactly was the condition observed?
- TIME - How recent is the report?
- ROAD CONDITION - Can the road be passed?
- SEVERITY - How serious is the reported flooding?
- STATUS - Has the report been reviewed, updated, or cleared?
- TRUST - Is the report unverified, reviewed, or supported by multiple/verified reports?

Instead of searching through scattered posts, users can select a specific road and see its latest available reported condition, timestamp, verification status, supporting evidence, and condition history in one place.

EXAMPLE SITUATION

A student needs to travel through a particular road during heavy rainfall. Instead of checking several group chats and social media posts, the student opens BAHAWATCH and selects the road.

The map shows:

AREA A

- Flood Severity: High
- Road Condition: Not Passable
- Reported: 8 minutes ago
- Photo/Video: Available
- Status: Verified
- Additional Report: 2 community reports

Later, a new report may show:

Flood Severity:

- Moderate
- Reported 5:20 PM
- Status: Updated.

The road's condition history can therefore show that the reported situation has changed over time.

MAIN FEATURES

- Flood & Road Condition Map - Shows available reports by location.
- Flood Report Form - Allows safe submission of a report.
- Automatic Location / Map Pin - Connects the report to a specific area.
- Road Condition - Shows whether the road is passable, difficult to pass, not passable, or cleared.
- Report Details - Shows severity, location, time, photo, and status.
- Freshness Indicator - Shows how recent the report is.
- Verification Status - Distinguishes submitted reports from reviewed/verified information.

- Optional Photo/Video - Users may attach a photo or short video showing the reported road condition. Media should only be captured from a safe location and should never require the user to approach floodwater or stop dangerously.
- Road Condition Timeline - Shows how the reported condition of a specific road changes over time based on available community reports.
- Admin Dashboard - Allows authorized reviewers to review, update, verify, or remove reports.

MVP

The Minimum Viable Product (MVP) of BAHAWATCH will focus on providing a simple, map-based app for viewing and submitting recent, location-specific reports on flood and road conditions. The MVP is designed to validate whether users find structured and time-stamped road condition information useful when assessing a specific area.

The MVP will consist of the following core components:

1. **Interactive Community Map**
   A map interface displaying reported locations and their corresponding road conditions. Users can select a reported location to view available information.

2. **Location-Specific Road Condition Reports**
   Each report will contain the reported location, road condition, severity level, date and time of submission, and an optional description. Road conditions may be categorized as Passable, Difficult to Pass, Not Passable, or Cleared.

3. **Photo and Video Evidence**
   Users may optionally attach a photo or short video to provide additional context regarding the reported road condition. Media submission will not be required, and users will be reminded to submit reports only from a safe location.

4. **Report Freshness and Timestamp**
   The app will clearly display when a report was submitted so that users can determine how recent the information is. Older reports will not automatically be treated as current conditions.

5. **Report Review and Verification**
   Submitted reports will be placed under review before being identified as verified. The app will distinguish between submitted, under-review, verified, needs-update, and cleared reports to provide users with appropriate context regarding the reliability of the information.

6. **Road Condition Timeline**
   Where multiple reports are available for the same location, the app will organize them chronologically to show how the reported road condition has changed over time. This feature will be tested with users before being treated as the main differentiator.

7. **Basic Administrative Dashboard**
   An administrative interface will allow authorized users to review submitted reports, examine supporting evidence, update report status, and manage reports that require verification or updating.

The MVP will intentionally exclude advanced features such as predictive flood modeling, large-scale analytics, automated emergency decision-making, full institutional monitoring systems, and integration with live weather broadcasts. These may be considered for future development after the core concept has been validated with target users.

POSSIBLE FUTURE FEATURES

- Flood History - Identify locations that frequently experience flooding after enough historical reports have been collected.
- Multiple Reports - Group reports referring to the same area.
- Community Confirmation - Allow users to confirm whether a condition is still present.
- LGU / DRRM Dashboard - Provide authorized institutions with monitoring tools.
- Area Alerts - Notify users about new reports in selected locations.
- Flood Data Analytics - Analyze accumulated reports to identify recurring patterns.
- Low-Connectivity Features - Explore options for areas with unreliable internet access.
- Wider Coverage - Expand to additional municipalities or cities after validation.

These are future possibilities, not requirements for the first MVP.

VALIDATION

Before assuming that people need the app, the team should validate the problem with actual potential users. The primary validation group is residents and students/commuters who regularly travel through flood-prone roads, with motorists, delivery riders, and businesses as additional groups. The team should also compare BAHAWATCH with the tools people already use, including social media, messaging platforms, navigation apps, and official disaster-apps.

Validation Questions

- How do you usually find out if a road is flooded?
- What source do you normally use?
- Have you ever encountered a flooded road without knowing beforehand?
- How difficult is it to find the condition of a specific road or area?
- What information do you look for before travelling?
- Do existing sources provide enough location-specific information?
- Would a road-specific page showing recent reports, freshness, verification status, and condition history be useful?
- What would make you trust a community report?
- Would you submit a report if you encountered flooding and were in a safe location?

The team should record actual responses and use the results to refine the MVP. No survey percentages should be presented until they have been collected. The initial validation should focus on the school community, while the same interviews can identify whether the problem also exists among wider communities and potential institutional customers.

WHAT WE NEED TO PROVE

- The existence of flooding is already known. The key hypothesis to validate is whether people have difficulty finding recent, location-specific information about actual road conditions.
- People may rely on scattered sources for local flood updates, but the extent of this problem must be measured through actual user validation.
- Existing sources do not always answer whether a specific road is passable.
- Users find a road-specific view of recent reports, freshness, verification, and condition history useful enough to check when deciding what information they need.
- Some users are willing to contribute reports when safe.
- Users need clear indicators of report freshness and trust.

USER ADOPTION STRATEGY

BAHAWATCH will not assume that people will abandon platforms they already use. The initial strategy is to focus on a narrow, high-value use case: quickly checking the latest reported condition of a specific road within and around the school community during flooding. The team will test this with actual users and identify what information is missing from their current sources.

The pilot should begin with the school community as the initial validation market, where the team can recruit students, faculty/staff, and other campus users as initial contributors and testers. This limited scope will make it easier to collect enough reports to demonstrate the road-specific experience. After the problem, usage, and value are validated, BAHAWATCH can expand to nearby communities and eventually to LGUs/DRRM offices.

POSSIBLE WEAKNESSES

- Not everyone will report because they may not have internet access or may not be in a safe position to use a phone.
- Reports can become outdated because flood conditions change quickly.
- False or inaccurate reports are possible.
- Coverage depends on the availability of reports.
- Users need an internet connection for the mobile app MVP.

HOW WE CAN ADDRESS THE WEAKNESSES

Outdated Reports
Display exact submission time and freshness indicators, and allow reports to be marked as needing an update or cleared.

False Reports
Use administrator review, verification status, report flagging, timestamps, photos when safely available, and multiple reports as supporting evidence.

Lack of Reports
Clearly show when an area has no recent reports rather than assuming it is safe or clear. Explore partnerships with LGUs and community organizations as the app develops.

Safety
Require safe reporting behavior and avoid encouraging users to approach floodwater or use the app while driving.

Internet Problem

Keep the initial MVP web-based and prioritize reliable online functionality. Evaluate offline or low-connectivity support only after validating the core workflow.

Connectivity Limitation and Future Offline Support

As a mobile app, BAHAWATCH currently depends on network connectivity for accessing updated reports and submitting information to the central system. This is a known limitation, particularly in areas where mobile data or internet access may be unreliable during severe weather. The initial MVP will therefore prioritize reliable online functionality and clearly communicate the age of available reports rather than implying continuous coverage. Future development may introduce low-connectivity capabilities, such as locally storing prepared reports and synchronizing them when a connection becomes available, as well as caching previously loaded information with clear update indicators. These capabilities will be considered after the core reporting and road-condition workflow has been validated with users.

If there is NO network

For viewing:

- The app can show “previously loaded/cached reports” and their timestamps.
- It should clearly indicate that the information may be outdated.

For submitting:

- Let the user “prepare a report offline”.
- Save it temporarily on the device.
- Once the connection returns, the report is uploaded automatically.
- The report should retain the time it was actually created, rather than pretending it was submitted at the later upload time.

SUCCESS METRICS

- Number of validated users who test the MVP.
- Number of reports successfully submitted and displayed.
- Percentage of reports that can be reviewed or verified.
- Average age/freshness of reports available on the map.
- Percentage of test users who can find a specific road condition without assistance.
- User-rated usefulness and trust of the information.
- Number of locations with usable reports during the pilot.

POSSIBLE USERS VS. POSSIBLE CUSTOMERS

Initial Pilot Users
- Students and commuters within the school community
- Faculty, staff, and other campus users
- Nearby residents and road users
- Drivers and delivery riders
- Local businesses

Potential Institutional Partners / Customers
- School and institutional partners
- LGUs / DRRM offices
- Disaster response organizations
- Community organizations

- Other organizations involved in disaster preparedness

The school community will serve as the initial market for validation and pilot testing. The team will first determine whether users experience the information gap, use the app, and find the road-specific reports useful. Wider communities and institutional organizations will be treated as expansion markets or potential partners/customers only after the concept has been validated and the app shows sufficient value for their needs.

POSSIBLE STARTUP POTENTIAL

BAHAWATCH can begin with the school community as a focused pilot market and expand only after demonstrating a real information need, repeated use, user value, and trust in the reports. The initial community-facing app can remain free to encourage participation, while the business model can develop around paid institutional features and services for schools and, eventually, LGUs/DRRM offices.

A possible sustainability model is:

- Free community access for viewing and submitting reports.
- Paid institutional features for schools and, eventually, authorized LGUs/DRRM offices, such as dashboards, monitoring, report management, historical data, and analytics.
- Potential service or partnership arrangements for customized monitoring, analytics, and other institutional needs.

The first priority is not monetization at scale. The priority is proving:

- People experience the information gap
- People find the road-specific view useful
- People return to use it
- The information is useful and trusted
- The app can be maintained.

THE MAIN IDEA

- We are not trying to predict floods.
- We are not trying to replace weather applications.
- We are not trying to replace official disaster warnings.

We are solving a specific information problem:

When flooding happens, how can people more easily find recent information about the reported condition of a specific road or area?

Our answer is a road-focused community app that organizes reports by location, time, road condition, severity, freshness, and verification status, with a condition timeline for each road when enough reports are available.

THE CORE VALUE
Weather tells you what may happen. BAHAWATCH helps you quickly see what has recently been reported about the specific road you are checking.

Potential signature feature: Road Condition Timeline - a chronological view of available reports for one road, showing how its reported condition changes over time. This feature will be tested with users before being treated as the main differentiator.

GOOGLE FORM RESPONSES

1. What problem do you want to solve?
We want to solve the information gap between general weather or flood warnings and the actual reported condition of a specific road or area. During heavy rainfall, people may know that flooding is possible without knowing whether the route they need to use is currently flooded, passable, difficult to pass, or already cleared. They often rely on scattered Facebook posts, group chats, word of mouth, or other sources. BAHAWATCH organizes recent, time-stamped community reports in one map so users can more easily find location-specific information and judge how fresh and verified a report is.

2. Who are the people or groups affected by this?
The primary users are residents and students/commuters who need to check a specific road during flooding. Motorists, delivery riders, and local businesses are secondary users with related needs. LGUs

and disaster response groups are potential institutional partners because organized community reports may provide another local information source for monitoring, subject to verification and official procedures.

3. What is your proposed solution? (100 words)
We propose BAHAWATCH, a community-powered flood and road-condition reporting app that organizes recent reports for specific roads on an interactive map. Users can safely report a flooded road or area by providing its location, road condition, severity, submission time, and an optional photo/video. Reports show their freshness and review/verification status, while a road-specific timeline can show how reported conditions change over time. Unlike a general weather source, BAHAWATCH focuses on organizing recent community-reported ground conditions for the specific road a user is checking. It does not predict floods, replace official advisories, or tell users which route to take; it helps them understand the latest available reported condition.

4. Which focus area?
Climate & Environmental Sustainability

5. What stage is your idea currently in?
Idea/Concept only

6. What do you envision as your MVP?
Our MVP will be a focused mobile app for road-condition viewing and reporting. Users will be able to select a specific road, view its latest available report, freshness, verification status, and condition timeline, and submit a new report when safely located. Each report will contain a map location, road condition, severity, timestamp, and optional photo/video. A basic admin dashboard will allow authorized reviewers to check, verify, update, clear, or remove reports. The prototype does not need to connect to a live weather broadcast yet; it will focus on demonstrating the core reporting and road-condition workflow.

7. Who are your target users or customers?
Our initial users are students, faculty/staff, and other members of the school community who need to check the condition of specific roads during flooding. Nearby residents, motorists, delivery riders, and local businesses can become additional users as coverage expands. Potential institutional partners or customers include schools, LGUs/DRRM offices, disaster response organizations, and community organizations. We will validate the school community first, then determine which wider user groups and institutions experience sufficient value to support expansion.

8. How will your solution create value?
BAHAWATCH creates value by organizing recent, road-specific ground reports into one easy-to-check view. Instead of searching through scattered posts or chats, users can select a particular road and see the latest reported condition, severity, submission time, verification status, and condition history when available. This does not replace existing platforms; it gives users a focused way to find, compare, and understand scattered local observations. For institutions, an organized collection of community reports may provide another local information source for monitoring affected areas, subject to verification and existing procedures.

9. Do you see potential for this idea to become a startup? Why?
Yes, subject to validation. The app can start with the school community as a focused pilot and expand only after proving a real information need, repeated use, user value, and reliability of reports. A future institutional version could provide schools and authorized LGUs or organizations with dashboards, monitoring, report management, historical data, and analytics. Community access can remain free while institutional services can provide a sustainability model. The key is to validate the problem first, test usage and willingness to pay, improve the product based on evidence, and then expand coverage.

10. What skills does your team have?
Programming / Software Development; UI/UX / Design; Entrepreneurship; Marketing / Communications; Research / Data Analysis.

11. Why should your team be selected as one of the Top 10?
Our team combines the technical, design, research, communication, and entrepreneurship skills needed to move this idea from validation to a working MVP. We can investigate how people currently find road-condition information, compare the concept with existing platforms, design a focused reporting experience, develop the map and database workflow, analyze feedback, and communicate the product's value. More importantly, we are prepared to test our assumptions and refine the solution based on evidence rather than claiming innovation without validation. Our goal is to produce a practical, demonstrable MVP that addresses a real community information problem.

12. What do you hope to achieve through CODELYMPICS 2026?
We hope to turn BAHAWATCH from a concept into a tested working MVP. Through mentoring and validation, we want to determine whether users genuinely experience difficulty finding the latest condition of a specific road, what they currently use instead, which report details they trust, whether the road-specific timeline is useful, and what would make them return to the app. We also want to strengthen our technical, business, research, and presentation skills and develop a realistic path for continuing the MVP after the hackathon.

13. If selected, are you willing to continue developing your MVP after the hackathon through the programs and mentoring activities of ISATech Society and KWADRA TBI?
Yes.

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